Nexus
Research article

Desert power for the AI era

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IN BRIEF

Zhu et al. investigated a sustainable pathway for the AI era by co-locating data centers with off-grid wind, solar, and storage systems in desert regions. Addressing the energy-computing nexus, this study demonstrates that transmitting data rather than electricity offers a potentially viable solution to the growing computational energy bottleneck.

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BROADER CONTEXT

The rapid expansion of artificial intelligence (AI) poses emerging challenges to electricity supply and climate goals. Addressing this critical energy-computing nexus, we investigate a potential solution: co-locating data centers with off-grid wind-solar-storage systems in desert regions. By utilizing efficient fiberoptic data transmission to circumvent the challenges of long-distance power transport, this approach offers a possible pathway to convert renewable resources into computational power. Our analysis suggests that this strategy could help meet growing AI energy demands in an economically viable and environmentally sustainable manner.

ABSTRACT

The surge in generative artificial intelligence (AI) may cause growing conflicts between securing electricity supplies and achieving sustainable development goals. Here, we propose off-grid hybrid wind-solar-storage (WSS) systems, which leverage the immense renewable resources in desert areas alongside relatively low-cost fiberoptic connectivity of data centers, to address this challenge. Using a global high-resolution techno-economic optimization model, we demonstrate that well-planned WSS systems can deliver cost-effective 24/7 uninterrupted power, primarily tailored for energy-intensive, latency-tolerant foundation model training. Furthermore, our analysis reveals that regions proximate to load centers can also support latency-sensitive inference tasks. This energy supply is capable of delivering 1 PWh globally in 2030 at levelized costs of around \$39/MWh, meeting the forecasted electricity demand for AI by the International Energy Agency. Moreover, even if AI electricity demand increases 10-fold, reaching 10 PWh by 2030, the unit cost increment would be less than 20%. Further uncertainty analysis shows that under extreme investment (3.0×) and cooling (2.0×) cost assumptions for data centers operating with the desert WSS systems, this 10-PWh/yr demand could still be satisfied at competitive cost levels. The desert WSS systems could potentially align computational and clean energy infrastructure in the AI era, as well as simultaneously achieving decarbonization and ecological restoration.

INTRODUCTION

The advent of generative artificial intelligence (AI), particularly large language models (LLMs) such as ChatGPT, has been applied unprecedentedly across human society and is widely regarded as a pivotal technology for the next wave of scientific and industrial progress. This acceleration intensified with the 2025 introduction of DeepSeek-R1,[1] a highly capable open-source reasoning model that, together with other high-performance open-source models[2,3] and application fusion, democratizes access to advanced AI capabilities globally.[4] These developments have precipitated a surge in computational demand: AI systems currently consume over 400 TWh of electricity annually worldwide,[5] with platforms like ChatGPT processing over 1 billion queries daily,[6] yet this represents merely the beginning of an exponential trajectory.[7] Current estimates suggest that training a single LLM equivalent to GPT-3 requires approximately 1.3 GWh of electricity.[8] Projections indicate that the overall electricity demand might increase to nearly 1,000 TWh/yr (1 PWh/yr) by 2030[5,9] as AI becomes ubiquitous across society.[10–13] However, this rapid proliferation of increasingly sophisticated models may create a critical bottleneck in securing adequate electricity supplies to power the computational infrastructure essential for continued AI development. This escalating energy demand poses significant challenges to sustainable development goals (SDGs),[7] as some proposals advocate for reactivating fossil fuel plants to meet immediate power requirements[14]—a direction that contradicts global commitments to net-zero.

Although large-scale renewable energy deployment offers a solution to this paradox between high electricity demand[15] for AI and SDGs,[16] it confronts significant constraints in most load center areas, including land availability,[17] electricity use competition with other sectors (e.g., electricity for basic needs), delays in grid interconnection approval,[18] and generally unfavorable wind or solar photovoltaic (PV) resources. Desert regions, covering over 20% of the Earth's land surface,[19] present an untapped opportunity excluded from traditional development considerations. While previous proposals for desert solar facilities—such as Sahara-to-Europe transmission schemes—have faced difficulties due to prohibitive electricity transmission costs,[20] this perspective may overlook a critical paradigm shift enabled by data centers: unlike electricity or hydrogen, which depend on expensive physical transport infrastructure, data transmits instantaneously through less costly fiberoptics and satellite systems. This creates potential synergy: desert regions can leverage abundant renewable resources and vast available land while circumventing traditional transport barriers through digital outputs (see the conceptual schematic in Figure 1). Furthermore, the remoteness of desert sites mitigates the impact of noise and thermal pollution on densely populated areas, issues often associated with hyperscale computing facilities. Co-locating energy generation with data centers eliminates transmission losses, offering a viable pathway that simultaneously addresses AI's electricity bottleneck and capitalizes on underutilized desert resources. Recent real-world pilot studies have suggested the viability of this approach.[21]

However, realizing this vision requires overcoming critical challenges in resource assessment, off-grid system reliability, and economic viability. First, the resource potential and spatial distribution of suitable desert locations need to be comprehensively assessed. This is necessary to identify sites capable of supporting large-scale renewable energy installations while satisfying stringent data center operational requirements, particularly maintaining suitable heat dissipation conditions essential for server performance. Second, the primary technical obstacle lies in developing robust off-grid power systems that can provide an uninterrupted electricity supply. Although the lack of grid connectivity in most desert regions forces these facilities to be fully autonomous, this mandatory off-grid configuration potentially offers strategic advantages: it guarantees 100% renewable energy consumption, insulates public utility grids from the severe load-ramping characteristic of large-scale model

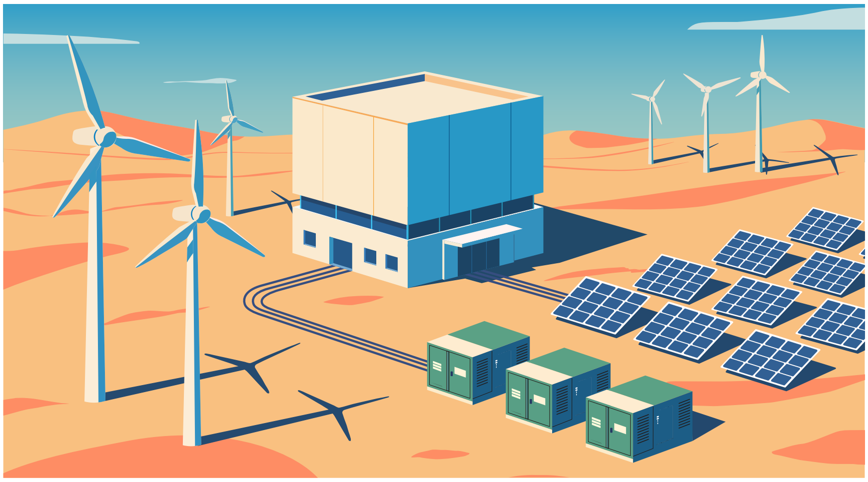

Figure 1. Conceptual schematic of a desert-based, off-grid hybrid wind, solar, and storage (WSS) system designed to meet the power demands of the AI data center

training,[22] and circumvents the protracted timelines associated with grid interconnection approvals to match the rapid pace of AI deployment. Consequently, advanced energy storage systems must be designed to handle not only diurnal solar variations but also seasonal wind patterns and potential extreme weather events that could affect renewable generation for extended periods.[23] Furthermore, the economic viability of establishing desert AI data centers (AIDCs) must be rigorously evaluated to determine whether this approach represents a cost-effective and long-term sustainable solution. Therefore, quantifying the technical and economic development potential, alongside spatial distribution patterns of this approach, holds profound significance for creating advanced computational infrastructure for artificial general intelligence.

Here, by developing a high-resolution techno-economic optimization model, we systematically assess the potential of this system at the global scale. First, we collect geographic data (e.g., land use, slope, elevation) and meteorological data (e.g., solar radiation, ambient temperature, wind speed) to evaluate the installation potential and hourly generation profiles for wind and solar power across global desert regions. Second, we apply the environmental criteria necessary for data center construction to exclude unsuitable desert areas, such as those with excessively high ambient temperatures. Third, we develop a model with high spatiotemporal resolution to identify the optimal capacity mix of wind, solar, and battery storage needed for over 40,000 grid cells to meet a certain given power output profile, such as uniform 1-GW output for all 8,760 h of the year. Model training serves as the ideal primary application scenario for these remote desert AIDCs, primarily due to its inherent latency tolerance. Furthermore, the energy consumption during this training phase is significantly higher than that of the inference phase. Therefore, we assume all data centers require a stable 24/7 power supply to simulate the rigorous demands of training, thereby avoiding any overestimation of power supply potential. Nonetheless, we also explicitly evaluate the geographical subset of these resources that satisfies the low-latency requirements for real-time inference, thereby providing a tiered assessment of global potential. Based on the resource potential data and the grid-cell-level ($0.25^\circ \times 0.25^\circ$) wind-solar-storage (WSS) systems optimized by the model, we evaluate the techno-economic electricity supply potential for the AIDCs in the near future (2030). Our results indicate that constructing large-scale WSS systems in deserts to support AI-focused data centers is economically viable, capable of supplying 10 PWh/yr electricity for AI, equivalent to the training of over 7 million LLMs comparable to GPT-3, at a reasonably low power cost of \$32–\$60/MWh.

RESULTS

Wind and solar resource in global desert area

The viability of our proposed desert-based AI data center strategy is anchored in the exceptional renewable energy endowment available across global desert regions. These areas, covering over 20% of the Earth's land surface, represent a vast frontier for data center deployment. Our comprehensive assessment shows that if this potential were fully utilized, the combined wind and solar generation exceeds 1,330 PWh annually, a figure representing over 50 times the global electricity consumption in 2020.[24] This extraordinary concentration of energy resources provides the essential foundation for economically viable off-grid AI data center operations, offering a scalable solution to meet the escalating energy demands of the AI era.

Beyond sheer scale, the quality of these desert renewable resources is world class, substantially exceeding global averages. For wind power, prime locations are concentrated across North Africa's Saharan margins, Central Asia's expansive steppes, and western North America's Great Basin, where annual average capacity factors often exceed 0.4 (Figure 2A). These numbers are comparable to those of the most productive offshore wind farms globally. The solar resource also demonstrates high potential, with 75% of the assessed desert areas' capacity factors exceeding 0.22 (Figure 2B)—more than double the global solar PV average of about 0.11.[27] This combination of high-quality wind and solar resources underscores the opportunity presented by desert regions.

A critical advantage of desert locations for data centers lies in their strategic proximity to existing global telecommunications networks. While prohibitive electricity transmission costs have historically left these energy-rich regions underutilized, data centers circumvent this barrier. The submarine fiberoptic cables that form the intricate backbone of modern data transmission land extensively on coastlines adjacent to major desert systems (Figure 2). This reduces the challenge of data connectivity to manageable terrestrial fiber extensions from coastal submarine cable landing sites to inland data center sites (e.g., internet exchange points [IXPs]). The economic implications are profound: this fiberoptic infrastructure in a desert area costs merely tens of thousands of dollars per km,[28] in stark contrast to electrical transmission infrastructure, which can account for several million dollars per km.[29] This economic disparity reinforces our approach that relocating computation to energy production centers is vastly more efficient than transmitting energy to population centers, transforming a perceived geographic disadvantage into a strategic opportunity for sustainable computational infrastructure.

Combination of wind, solar, and storage for stable power output

While the potential is substantial, reliance on renewable energy resources traditionally raises concerns about power supply stability and intermittence[29] as data centers require uninterrupted 24/7 operations. In our model, this standard is technically defined as achieving 100% temporal availability through the optimized dispatch of wind, solar, and battery storage. Furthermore, a 5% operational reserve margin is strictly maintained to buffer against supply-side variability and to ensure reliability even under adverse conditions or extreme weather events. Previous work has demonstrated that the well-planned deployment of variable renewable energy and storage could satisfy the load demand with any specific profiles.[30] Motivated by this, we develop an optimization model to identify the capacity mix of wind, solar, and storage required in each grid cell to deliver a constant electricity

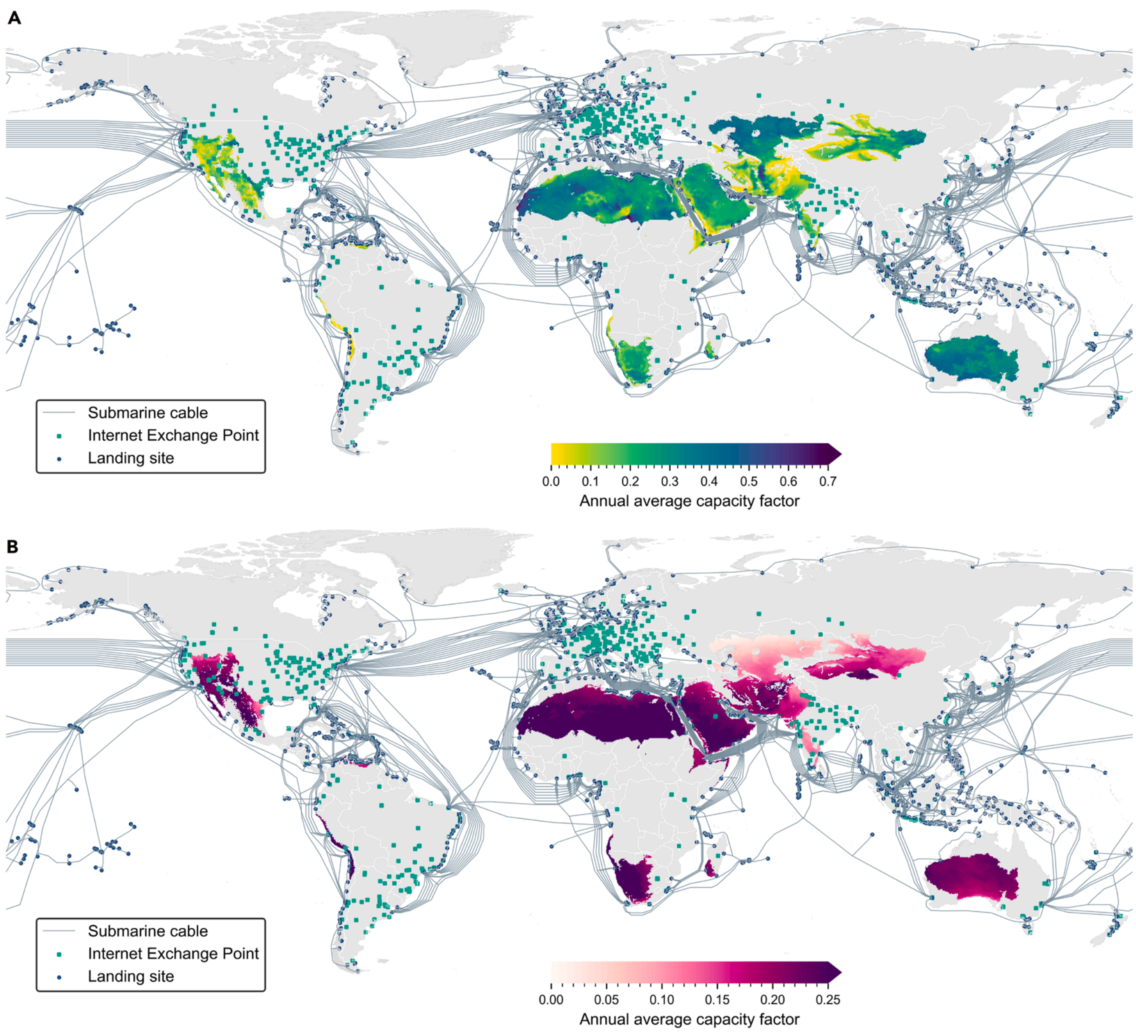

Figure 2. Wind and solar resource potential in global desert regions

(A and B) Annual capacity factor (0–1) for wind (A) and solar PV (B). The spatial resolution for wind and solar resource is at 0.25° × 0.25° with hourly assessment reanalysis meteorological data from ERA5.[25] The network infrastructure data is derived from PeeringDB.[26]

output for all 8,760 h of the year. This model illustrates that a tailored combination of wind, solar, and battery storage can mitigate the inherent variability of individual renewable resources, thereby achieving a stable power supply that meets the operational demands of AIDCs. Figure 3A (A1–A3) shows the distribution of wind, solar, and storage combination across the global desert area. We find that the optimal capacity mix exhibits profound spatial heterogeneity, reflecting the quality and complementarity of local wind and solar resources. This spatial variation underscores the necessity of a location-specific approach to system design, ensuring both reliability and economic efficiency.

We show the operational dynamics of a typical WSS system in the representative grid cell in Figures 3B and 3C. This representative grid cell is selected with the minimum total capacity of wind, solar, and storage optimized by our combination model for stable power output (e.g., 1 GW) across all grid cells. This operation profile demonstrates the seamless integration of these technologies to maintain a constant power supply. Throughout a full year, the system adeptly balances the fluctuating outputs of wind and solar generation. During periods of high renewable output, surplus energy is used to charge the battery storage system; during deficits, the stored energy is discharged to fill the gap, ensuring the power supplied to the data center remains stable. This dynamic interplay resolves on a daily and seasonal basis (Figure 3C). The typical daily profile shows solar generation peaking midday, while wind power may be more prominent at night, with storage acting as the critical link that smooths these variations. Across seasons, the system adapts to shifts in weather patterns, such as lower solar insolation or changing wind regimes, by adjusting the dispatch from each component to guarantee unwavering reliability. This demonstrates a robust and resilient power solution capable of supporting critical AI infrastructure off grid.

Configurations for stable output vary significantly by geographic location, as illustrated by the optimal capacity ratios for wind, solar PV, and storage (Figures 3A1–A3). A key finding is that regions with strong, complementary diurnal and seasonal patterns of both wind and solar resources require a smaller total installed capacity. For instance, areas in the Sahara and Middle Eastern deserts, which benefit from both high solar irradiance and consistent wind regimes, can achieve stable output with a relatively balanced and lower overall capacity. Conversely, regions where resources are imbalanced–such as areas in Central Asia with world-class wind but less abundant solar resources, or other areas with excellent sun but poor wind—necessitate a larger installed capacity of the dominant resource, along with greater storage capacity, to compensate for the intermittent supply and ensure continuous power generation. The distribution of these diverse capacity combinations is further detailed in Figure S11, highlighting the spectrum of optimization strategies required to harness desert renewables effectively.

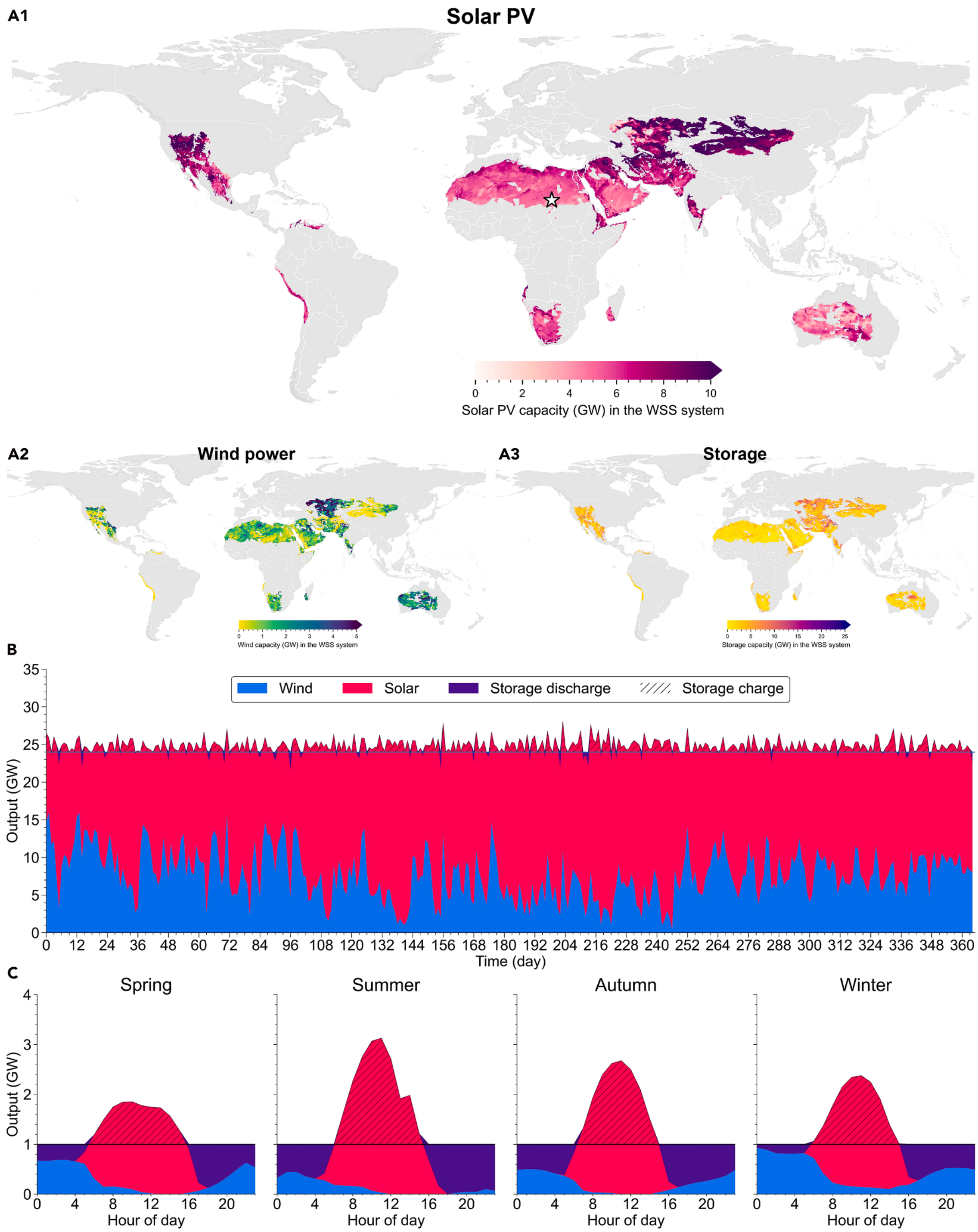

Figure 3. Configuration of WSS systems for desert data centers

(A1–A3) Capacity mix of wind, solar PV, and storage for stable power output (1 GW) in each desert grid cell.
(B) Daily power generation and stable power output (1 GW × 24 h) in a sample grid cell (black star in A1).
(C) Hourly power generation profile for 1-GW stable output in typical days across seasons.

Environmental suitability, site selection, and latency

Two primary concerns have traditionally deterred data center deployment in desert areas, despite these regions demonstrating exceptional renewable energy resources. First, elevated ambient temperatures in desert environments raise concerns about thermal management challenges, potentially increasing cooling requirements and associated operational costs. Second, the remote nature of many desert locations presents accessibility challenges that may complicate construction logistics, equipment transportation, and ongoing maintenance operations, potentially inflating both capital and operational expenditures.

To address these concerns, we first analyze global temperature patterns using 2020 meteorological data extracted from the European Center for Medium-Range Weather Forecasts Reanalysis V5 (ERA5) dataset[25] across all 0.25° × 0.25° grid cells worldwide. We find that the ambient temperatures in desert areas are milder than commonly

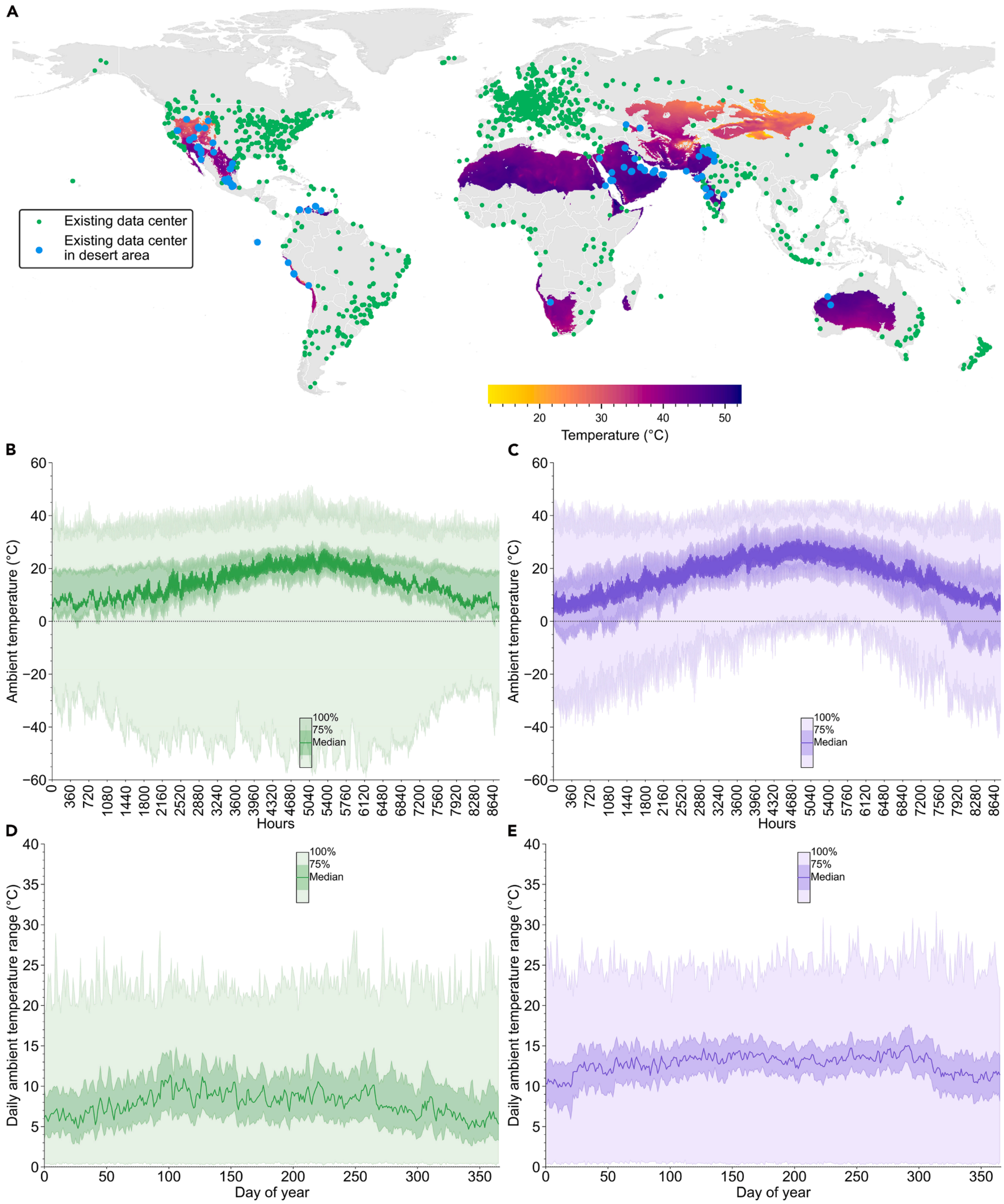

Figure 4. Ambient temperature (2020)
(A) Annual mean ambient temperature (°C) in global desert areas and location of existing data centers.
(B–E) Hourly distribution of ambient temperature (°C, B and C) and daily temperature range (°C, D and E) in existing data centers (B and D) and the cells meet the temperature filter standards (C and E). The existing data center location information is derived from PeeringDB.[26] The hourly temperature data are derived from the ERA5 dataset.[25]

perceived (Figure 4A), especially when compared with the existing data centers constructed by 2025 reported in PeeringDB,[26] which provides critical insights into operational temperature ranges. Analysis of annual environmental temperature data from these 5,200 operational data centers illustrates that the peak average ambient temperature approaches 30°C, with the absolute maximum ambient temperature reaching nearly 50°C (Figures 4B, S4, and S12). These operational precedents demonstrate that data centers can function effectively even under elevated ambient temperature conditions. Our comparative assessment of desert temperature conditions indicates that over 75% of desert areas experience maximum temperatures below 45°C, and 80% maintain temperatures below 30°C for at least 75% of the year. These findings show that the thermal conditions of most deserts are not extreme and can accommodate data center operations within established industry parameters. Notably, while isolated examples of successful desert data center operations exist, indicated by blue markers, the vast majority of current facilities remain concentrated in temperate regions, leaving desert areas largely undeveloped despite their demonstrated resource potential and thermal feasibility.

Beyond evaluating ambient temperature thresholds, we further assessed the operational energy efficiency of potential desert sites by modeling the annual power usage effectiveness (PUE) under a direct air-cooling regime. Leveraging hourly meteorological reanalysis data from ERA5—specifically incorporating dry-bulb temperature, dew point temperature, and atmospheric pressure—we derived PUE estimates across the identified regions. See detailed estimate methods in section S1.5 and results in Figure S10. The analysis reveals that the projected cooling energy overhead in these desert environments is not as prohibitive as traditionally assumed. While instantaneous PUE values may rise during intervals of peak thermal stress, the annualized metrics demonstrate that a significant proportion of these regions can maintain average PUE levels within a competitive range (most sites see average PUE less than 1.6). These findings suggest that when viewed through the lens of annual average performance rather than isolated peak extremes, the efficiency penalty of air cooling in selected desert areas remains within technically and economically manageable limits.

We adopt stringent temperature criteria to avoid possibly overestimating the potential, according to the ambient temperature of existing data centers and established industry recommendations, to exclude those grid cells with temperatures too high for data center operation. Specifically, we exclude grid cells where the annual mean temperature exceeds 25°C or the maximum temperature surpasses 46°C, a criterion aligning with International Business Machines.[31] The excluded areas are shown as gray regions with black dashed lines in Figure 5A. We further compare the temperature profiles of existing data centers with those of potential desert grid cells that meet the established thresholds. As shown in Figure 4C, we find that, overall, temperatures in desert regions satisfying the maximum temperature constraint do not differ too much from those at existing data center sites, while instances of extreme low temperatures (e.g., below −40°C) are substantially reduced. Additionally, we analyze the daily temperature range; as illustrated in Figures 4D and 4E, the median temperature range for existing data centers falls between 6°C and 8°C, whereas the median range in desert regions lies between 9°C and 16°C.

Furthermore, to evaluate long-term feasibility under climate change, we analyzed the temperature evolution of the selected desert regions from 2021 to 2100 using Community Earth System Model simulations under four shared socioeconomic pathways (SSP).[33] As illustrated in Figure S14, the annual mean temperature exhibits an upward trend across all scenarios, with warming rates of 0.011°C/year (SSP1-2.6), 0.033°C/year (SSP2-4.5), 0.054°C/year (SSP3-7.0), and 0.078°C/year (SSP5-8.5). Notably, even under the high-emission scenario (SSP5-8.5), the projected spatially averaged annual mean temperature reaches approximately 23.6°C by 2100, which remains below our strict exclusion threshold of 25°C. This indicates that the selected sites are climatologically robust and likely to sustain suitable thermal conditions for data center operations.

Even under stringent temperature constraints, we find the installed capacity potential for stable power output remains remarkably substantial in global desert regions (Figure 5B). The installed capacity potential for each grid cell is estimated as the maximum level of stable power output achieved by proportionally co-expanding wind and solar installations until the resource potential of either technology is reached. Formally, the potential (noted as $\overline{cap}_{wss}$) is equal to $\min\left(\frac{\overline{cap}_{we}}{cap_{we}}, \frac{\overline{cap}_{pv}}{cap_{pv}}\right)$, where $\overline{cap}_{we}$ and $\overline{cap}_{pv}$ are the resource potential in each grid cell, and cap_{we} and cap_{pv} are the capacity ratio optimized by the combination model, respectively. Globally, the aggregate WSS systems potential reaches an estimated 28,000 GW, a number equivalent to 28,000 times the capacity of the 1-GW super data center planned by OpenAI in the United Arab Emirates.[21] The vast majority of this potential is geographically concentrated in the Sahara Desert (about 11,800 GW) and the Taklimakan Desert in China (1,216 GW). Other critical regions include the Alashan Plateau (892 GW), the Central Persian Desert (583 GW), the Kalahari Xeric Savanna (448 GW), the Great Victoria Desert (398 GW), the Central Asian Southern Desert (370 GW), the Atacama Desert (366 GW), the Simpson Desert (347 GW), the Eastern Gobi Desert Steppe (319 GW), the Central Asian northern desert (314 GW), the Kazakh semi-desert (271 GW), the Arabian Desert (250 GW), the Gariep Karoo (244 GW), and the Junggar Basin semi-desert (230 GW).

We further examine transportation accessibility by matching the distance from the center of each grid cell that meets our temperature standards to the nearest established road infrastructure, utilizing global road data from the Global Roads Inventory Project (GRIP).[32] Beyond its promising scale, the identified 28,000 GW of WSS systems' potential is also highly accessible. We find that this resource is not geographically isolated; approximately 42.5% of the capacity (12,000 GW) is located within 30 km of major road networks, specifically those classified as secondary roads and above in the GRIP dataset (Figure 5C). This distance remains well within reasonable transportation ranges for construction equipment and maintenance vehicles. The accessibility is even more pronounced for a core fraction of this capacity: over 21% of the total potential (about 6,000 GW) is situated less than 10 km from the nearest major road. This widespread proximity to transportation corridors substantially mitigates a primary logistical and financial barrier to development, thereby enhancing the practical viability of harnessing these vast renewable energy resources.

To estimate the transmission latency for data centers in desert regions, we compile a comprehensive dataset of existing IXPs and submarine cable landing stations (see these points in Figure 2) and calculate the geodesic distance from each desert grid cell to the nearest connection point within the same country (or the nearest available point for the few African cells lacking domestic connections). Based on this geographic distance, we apply a fiber routing factor of 1.8[34] to conservatively estimate the physical fiber length and assume a data propagation speed of two-thirds the speed of light (approximately 200 km/ms) to calculate the round-trip latency,[35] incorporating an additional 5-ms overhead for equipment processing. Figure 5E illustrates the relationship between cumulative power generation potential and latency for grid cells that meet temperature constraints. We observe that the latency for all eligible desert grid cells remains below 50 ms; notably, the annual power generation potential reaches 160 PWh for locations with a latency under 15 ms and 76 PWh for those under 10 ms, demonstrating that with strategic planning, desert regions can effectively support not only AI model training but also real-time inference services. See the grid-cell level latency for those that meet temperature standards in Figure S13.

Economic viability and cost-benefit analysis

To evaluate the economic feasibility of our AIDC proposal, we calculate the levelized cost of electricity (LCOE) for WSS systems in the desert area. Our analysis initiates with current (2024) regional capital expenditure (CapEx) for wind, solar, and battery storage (see Table S6) and projects them to 2030 under three distinct technological advancement scenarios: rapid (CapEx-low), moderate (CapEx-mid), and slow (CapEx-high) (see Table S7). Significantly, our calculated WSS systems LCOE reflects the cost of guaranteed, stable power delivery—a metric fundamentally distinct from the LCOE of intermittent renewable sources alone. The detailed formulation is available in the section S1.6.

Figure 5A illustrates the LCOE for WSS systems within desert regions that satisfy our predefined thermal criteria. The results from our high-spatiotemporal-resolution assessment reveal a substantial global potential of approximately 10,000 GW of WSS systems capacity capable of providing firm power for less than \$68/MWh under our 2030 moderate cost-reduction scenario (CapEx-mid). These high-potential, low-cost regions are predominantly concentrated in China, North and Southern Africa, Australia, South America, and parts of the United States. When the acceptable LCOE threshold increases to \$100/MWh, the viable potential expands to over 25,000 GW, encompassing more than 90% of the total identified potential. By constructing supply curves that aggregate power potential in ascending order of LCOE (Figure 5D), we determine the cost of meeting specific demand

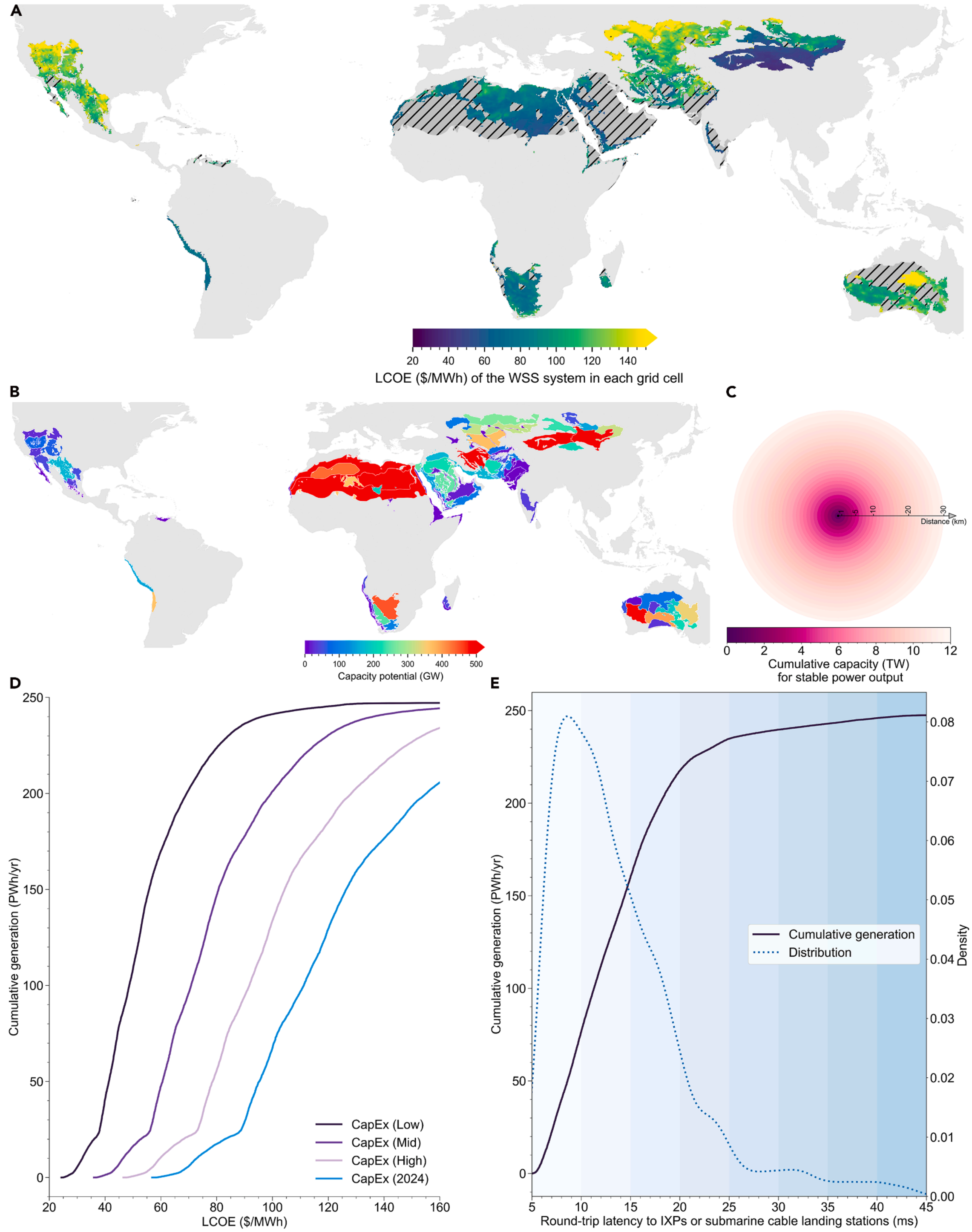

Figure 5. Potential and key features in the desert area for AIDC

(A) LCOE ($/MWh) for the WSS systems with stable power output in the grid cells meeting temperature standards. The gray region with black dashed lines shows grid cells that do not meet our temperature standards for data center operation.

(B) Installed capacity potential (GW) for stable power output at the desert-cluster level.

(C) The cumulative capacity potential (TW) against the distance from the cell's center to the nearest road. The global road network is derived from GRIP.[32]

(D) Cumulative electricity generation potential (PWh/yr) to the LCOE of the WSS systems in cells meeting standards.

(E) Cumulative electricity generation potential (PWh/yr) and kernel distribution to the round-trip latency to IXPs or submarine cable landing stations.

levels for AIDCs. In the CapEx-mid scenario, desert WSS systems meet the International Energy Agency's (IEA's) projected 2030 global electricity demand for AI (1 PWh/yr) at a remarkably low cost of \$39/MWh. Even if AI's energy requirement grows 10-fold to 10 PWh/yr—an amount equivalent to the total electricity consumption of China in 2024—the WSS systems can satisfy this demand at a cost of \$46/MWh. Across our three 2030 projected cost reduction scenarios (CapEx-low, -mid, and -high), the cost to meet the 1-PWh/yr demand ranges from \$27 to \$51/MWh, and for 10 PWh/yr, it ranges from \$32 to \$60/MWh. However, in an extreme bounding scenario where capital

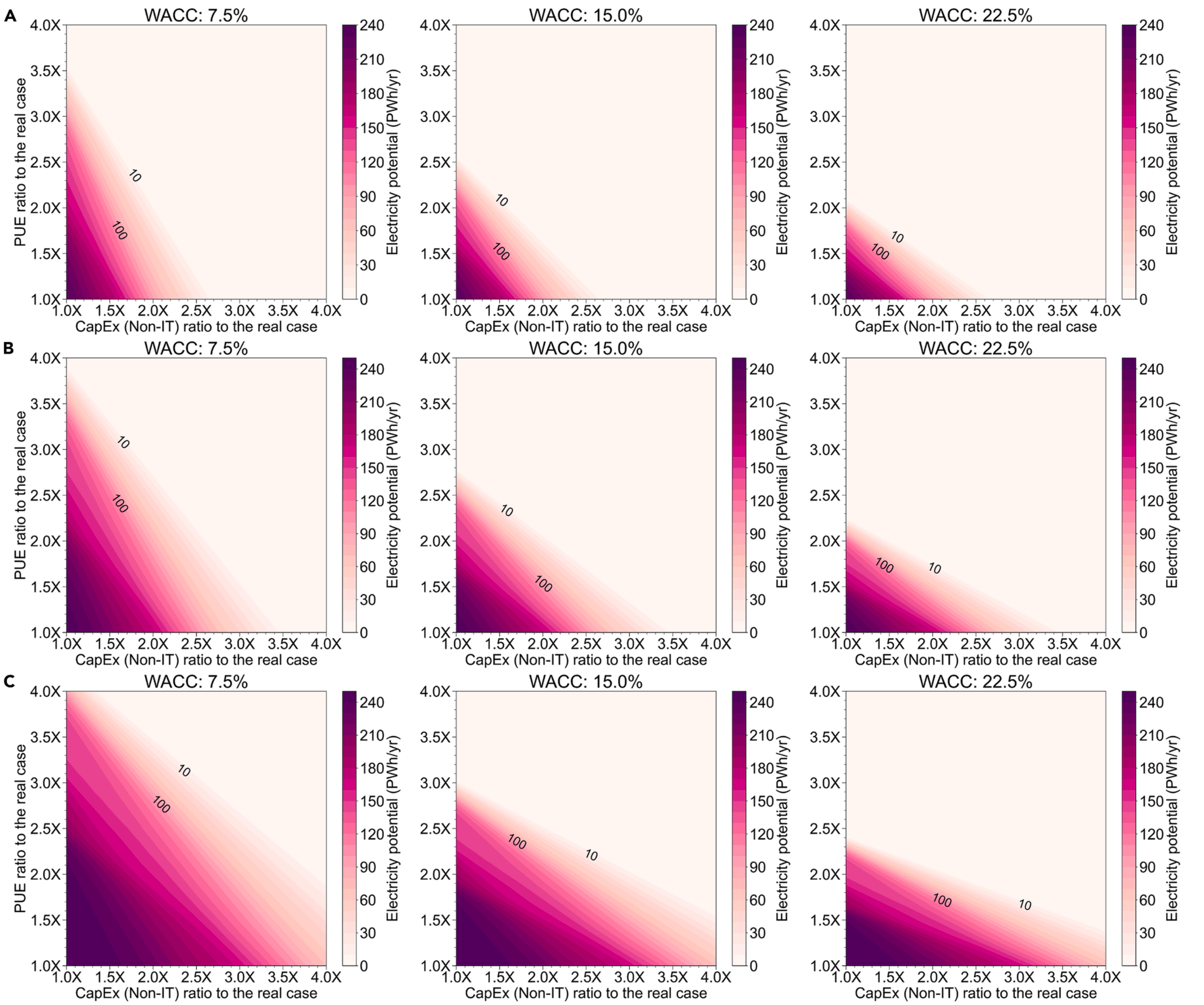

Figure 6. Economic potential of the WSS systems for AIDC in the desert area under diverse renewable investment cost in 2030
(A) CapEx-high; (B) CapEx-mid; (C) CapEx-low scenarios. The global economic potential (PWh/yr, color scale) is plotted against relative non-IT capital expenditure (X axis) and PUE (Y axis) to the reference case. Columns represent three financing scenarios with different WACC: 7.5%, 15.0%, and 22.5%. The key parameters on the real case data center are derived from Nanfang Wanguo Data Center.[36] The annual PUE value of the real reference case is about 1.2.

costs stagnate at 2024 levels, the cost to meet the 1-PWh/yr and 10-PWh/yr demands would peak at \$62/MWh and \$72/MWh, respectively.

The deployment of AIDCs operating powered by our proposed WSS systems in desert areas must overcome two principal challenges. The first is the incremental CapEx associated with constructing non-information technology (non-IT) infrastructure in harsh desert environments. The second is the increased operational cost needed to maintain IT equipment stability, which manifests as higher energy consumption by non-IT systems and can be reflected by an elevated PUE, the ratio of the total electricity consumed by the data center to the IT equipment ($\geq$1, lower is better). Given the real-world scarcity of directly comparable data centers—one conventional and one desert sited—we anchor our analysis to an actual project as a reference case (see parameters in the materials and methods section). To quantify the aforementioned negative impacts, we introduce multipliers for both the non-IT investment cost and the PUE relative to this reference, assuming consistent IT equipment investment costs in different operation environments (e.g., in desert areas or load centers). We then conduct a comprehensive parametric analysis, iterating through every thermally qualified desert cell to assess its economic potential under various CapEx scenarios of the WSS systems and weighted average cost of capital (WACC) requirements of the data center projects. For the WACC, we adopt values of 7.5%, 15.0%, and 22.5%, based on real-world projects, to represent a broad spectrum of investor return expectations relevant to data center projects. The primary objective of this analysis is to identify the boundary conditions for the non-IT CapEx and PUE multipliers that still permit a collective potential of at least 10 PWh of annual electricity supply across all scenarios. Through computational experimentation, we systematically vary these multipliers from 1.0$\times$ to 4.0$\times$ in increments of 0.01$\times$. This methodology creates a granular parameter space of 90,000 unique non-IT investment and PUE combinations for each cell, allowing for a robust assessment of economic viability. The specific equations are detailed in the materials and methods section.

Figure 6 presents the results of our economic potential assessment for deploying AIDCs in global desert regions under various 2030 renewable energy investment scenarios and WACC requirements. The parameter space below this line represents the portfolio of non-IT costs and PUE conditions that exceed this 10-PWh/yr threshold. Our results reveal that the potential for WSS systems to power AIDCs in desert regions remains substantial across a wide range of cost and financing

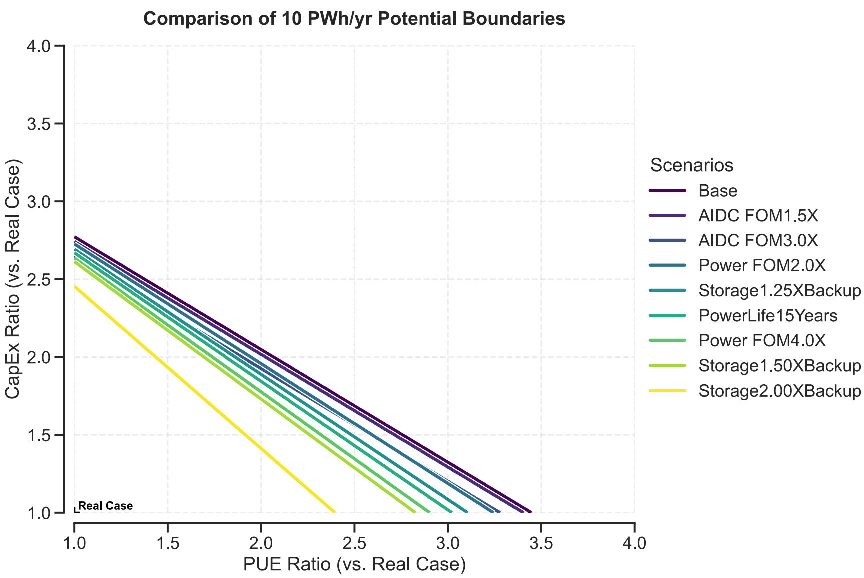

Figure 7. Comparison of PUE and non-IT DC CapEx boundaries for a 10-PWh/yr electricity potential across different scenarios
The plot illustrates the trade-off between the PUE ratio and the non-IT equipment CapEx ratio relative to the real case (marked at 1.0 and 1.0). Each contour line represents the specific combination of PUE and CapEx required to achieve an electricity potential of 10 PWh/yr. The base case here refers to CapEx-mid and WACC at 15.0% in Figure 6B. Scenario definitions are listed in Table S9.

assumptions. For instance, under the moderate 2030 cost scenario (CapEx-mid) and a 15.0% WACC, our analysis indicates that an AIDC can economically provide the target 10 PWh/yr of electricity as long as its PUE is below 2.8 times (absolute PUE value less than 3.36) that of the reference case and its non-IT infrastructure investment is less than 3.45 times the reference. Notably, even under the specific extreme conditions where non-IT equipment investment reaches 3.0×, and the PUE escalates to 2.0× relative to the reference case, our analysis confirms the system still robustly maintains the capacity to satisfy the 10-PWh/yr demand. This demonstrates a significant tolerance for the incremental costs associated with desert operations. We further test the potential of desert WSS systems to meet an extreme scenario of 100 PWh annual electricity demand by AI and found that this target could be achieved by keeping the PUE below 2.5 times and non-IT equipment investment costs below 2.8 times the values of our reference case under the CapEx-mid scenario for renewables' investment costs.

Cross-WACC comparison illustrates how this potential contracts as capital costs and perceived risks increase (left-to-right progression in Figure 6). Even under stringent financial conditions—characterized by a WACC of 22.5% and relatively high WSS systems investment assumptions (CapEx-high)—our analysis demonstrates that achieving 10 PWh of annual supply remains economically feasible, contingent upon PUE staying below 2.0× and data center non-IT CapEx remaining under 2.5× reference level. Under more moderate data center cost assumptions, where non-IT CapEx and PUE are limited to 1.8× above baseline scenarios, the economically viable potential expands dramatically to 100 PWh annually (Figure 6A). Our economic analysis confirms that the cost increase for AIDCs operating in desert regions could be compensated by the substantially reduced electricity prices available through co-located renewable generation. From a strategic market development perspective, our analysis maps a global supply curve for firm, clean, low-cost computational power by quantifying immense market depth available at various construction costs and supply chain conditions. This provides policymakers and corporate strategists with quantitative blueprints for cost-scale trade-offs, enabling long-range planning for the massive computational infrastructure required by the AI revolution.

Sensitivity analysis of system robustness

To assess the resilience of the proposed desert AIDC paradigm against the harsh environmental realities of desert deployment, we conduct a multidimensional sensitivity analysis focusing on the boundary conditions required to maintain a 10-PWh/yr global supply potential. Recognizing that desert environments may precipitate accelerated equipment degradation and elevated operational complexities, we introduce a series of conservative bounding scenarios explicitly designed to stress-test the economic resilience of the system. Rather than representing expected operational baselines, these stress tests deviate significantly from standard techno-economic assumptions to simulate extreme, worst-case operational environments. For instance, conditions such as hyper-arid zones with highly abrasive, persistent sandstorms could cause premature mechanical wear on turbine components or severe abrasion of PV modules, which we simulate by shortening the lifespan of wind and solar PV assets from the standard 25 years to 15 years ("PowerLife15Years"). Furthermore, the logistical challenges of highly remote locations lacking basic infrastructure could inflate service costs exponentially. To account for the necessity of continuous, intensive sand removal and complex remote logistics, we increase the fixed operation and maintenance (FOM) of non-IT equipment costs up to 3.0 times the base assumption (resulting in a cumulative 4.5 times the real-world benchmark). Similarly, the "Power FOM2.0X" and "Power FOM4.0X" scenarios are explicitly examined to simulate unprecedented operational burdens—such as the need for specialized robotic dry-cleaning fleets or premium labor rates in inhospitable climates—assuming the operation and maintenance (O&M) costs for wind, solar, and storage systems escalate to 2.0 and 4.0 times the base scenario levels, respectively. Finally, to buffer against catastrophic, prolonged extreme weather events (e.g., multi-day severe dust storms heavily occluding solar irradiance) while strictly maintaining off-grid reliability, we impose additional storage capacity redundancies ranging from 1.25 to 2.00 times the model-optimized requirement.

Our results demonstrate that while these extreme boundary conditions naturally contract the economically feasible solution space, the system maintains substantial viability (see economic potential space in Figure S15). This analysis reveals that the economic potential is most sensitive to the accelerated turnover of generation assets and extreme O&M cost escalations (see Figure 7). As illustrated by the "Storage2.00XBackup" and "Storage1.50XBackup" scenarios, the trade-off frontier between the allowable non-IT CapEx and PUE shifts downward compared to the base scenario. Similarly, the "Power FOM4.0X" scenario exerts a notable pressure on project economics by shifting the viability frontier; however, our data indicate that even with quadrupled maintenance costs for renewable assets, the 10-PWh/yr target remains achievable if non-IT construction costs are controlled within approximately 2.5 times the reference case. However, even under the most restrictive scenario—requiring twice the backup storage deployment—the system can still support 10 PWh/yr of electricity demand, provided that AIDC developers maintain reasonable efficiency standards (e.g., an annual PUE below 2.4) and limit non-IT construction costs to roughly double the reference case. This indicates that the sheer abundance and low cost of desert renewable resources provide a sufficient economic buffer to absorb the financial impact of significantly shortened equipment lifecycles and intensified maintenance regimes. Consequently, even if future desert AIDCs face compounding challenges—combining higher degradation rates, extreme maintenance requirements, and the need for massive storage buffers—the fundamental economic logic holds.

DISCUSSION

In this study, we systematically evaluated a novel paradigm for sustainably powering the AI era: co-locating AIDCs with dedicated off-grid WSS systems in global desert regions. Our high-resolution techno-economic analysis demonstrates that this approach provides a requisite 24/7 stable power supply while remaining economically robust across diverse financial and operational scenarios. By transforming costly long-distance electricity transmission challenges into manageable high-speed data exchanges, this model reveals immense, largely untapped renewable energy resources of global deserts to meet escalating AI computational demands. Specifically, this paradigm implies a globally distributed AI infrastructure hierarchy: remote, resource-rich deep deserts are ideally suited to serve as "global training factories" where massive foundation models are trained at the lowest levelized cost. In contrast, desert margins proximate to population centers can operate as "regional inference nodes," strategically balancing renewable energy costs with the low-latency requirements of end users.

Sustainable development co-benefits

Beyond direct economic advantages, the desert AIDC paradigm offers substantial co-benefits that strengthen its strategic value proposition. Primarily, carbon abatement represents the most immediate benefit: displacing fossil-fuel-powered grid electricity with renewable sources provides profound decarbonization potential.[29,37] As data center electricity consumption is projected to reach unprecedented levels in the coming decade, traditional grid-connected facilities relying on fossil-fuel-powered electricity would generate substantial carbon emissions. Desert AIDCs could achieve carbon neutrality during the operational phase through carbon-free WSS systems.

Ecological restoration may also represent an unexpected co-benefit with significant precedent.[38] Large-scale solar deployments may alter local land-surface properties through reduced surface wind speeds and water evaporation, facilitating soil stabilization and vegetation recovery.[39,40] Photovoltaic projects to combat desertification in China demonstrate practical applications,[41] and modeling studies suggest massive Sahara solar deployments could induce regional climatic feedbacks, potentially increasing local precipitation and promoting vegetation growth.[42] This creates positive feedback loops where clean energy generation simultaneously contributes to ecological restoration, transforming environmental liabilities into assets.

Additionally, economic development benefits extend to remote, often economically marginalized regions hosting these installations. Desert AIDC development could potentially create high-tech employment opportunities and economic value in areas traditionally excluded from the digital economy. For example, some infrastructure investments are increasingly accompanied by vocational training initiatives—exemplified by the Africa Data Center Talent Project and the World Bank's Digital Economy for Africa initiative—which aim to bridge the technical skills gap for local populations.[43] By aligning infrastructure deployment with workforce development, these projects not only foster regional economic resilience but also create a multiplier effect, where direct investment in data centers supports broader service-sector employment and technological capacity building.[44,45]

Implementation challenges and risk mitigation

Despite compelling advantages, the practical implementation of the desert AIDC faces critical challenges requiring proactive risk management strategies. Supply chain vulnerabilities represent immediate concerns, as wind turbines, solar PV panels, and battery manufacturing concentrate in limited geographic regions, creating potential bottlenecks and strategic dependencies. Geopolitical risks compound these challenges, with many resource-rich desert areas located in politically unstable regions. Concentrating critical global AI infrastructure in any single region would pose systemic risks. System security and data sovereignty concerns arise from remote installations in diverse political jurisdictions, potentially complicating regulatory compliance and data protection. A prudent mitigation strategy involves globally distributed deployment across multiple continents—spanning North America, Australia, Africa, and Central Asia—fostering resilient, decentralized networks that mitigate supply-side and geopolitical threats while ensuring distributed data security and sovereignty. This diversification approach transforms potential vulnerabilities into strengths by creating more abundant capacity and reducing dependence on any single region.

Water scarcity presents operational challenges in arid environments where traditional evaporative cooling technologies prove unsustainable. However, technological advancement rapidly addresses these constraints through emerging solutions, including direct-to-chip and full-immersion liquid cooling systems that drastically reduce or eliminate water consumption. While these advanced systems may require higher capital investments (reflected in the non-IT equipment investment cost multiplier) or altered energy consumption profiles, our economic framework demonstrates remarkable resilience to operational variations. Sensitivity analysis confirming economic viability even with PUE ratios reaching 4.0 indicates that profound cost reductions from locally generated renewable electricity create substantial financial headroom to absorb advanced low-water cooling technology costs.

Methodological contributions

Our study advances beyond existing literature in several critical methodological dimensions, as summarized in Table 1. Prior assessments of AI energy demand, such as IEA reports[5] and Goldman Sachs analyses,[46] have predominantly operated at regional or national aggregate scales with annual temporal resolution, focusing on macro-scale demand forecasting and investment opportunity assessment without coupling to physical renewable generation systems. Conceptual studies addressing AI sustainability challenges[47] have qualitatively discussed the energy-water-carbon nexus but lack quantitative optimization frameworks. Meanwhile, previous desert solar research[39,42] has examined climate feedback effects at relatively coarse spatial resolutions (several 100 km) with seasonal or monthly temporal granularity, typically considering single technologies under grid-connected assumptions. In contrast, our analysis uniquely integrates high-resolution spatial assessment ($0.25^\circ \times 0.25^\circ$ grid cells spanning over 40,000 cells globally) with hourly chronological dispatch optimization across 8,760 h annually, coupling WSS systems with data center thermal constraints and latency requirements to provide end-to-end techno-economic feasibility assessment for off-grid desert AIDCs with guaranteed 24/7 reliability.

Limitations and future directions

The objective of this study is to conduct a systematic, global-scale assessment of the techno-economic potential of the desert AIDC paradigm. Consequently, our analysis operates at a high level of abstraction, focusing on quantifying resource availability, system configuration requirements, and economic boundaries. We acknowledge several limitations inherent to this approach, which delineate critical pathways for future research.

First, while we incorporate elevated capital multipliers and operational expenditures to account for harsh desert conditions, we do not explicitly model granular engineering specifics such as the optimal choice of cooling technology or the precise logistics of construction. Desert regions present substantial operational challenges, including extreme diurnal temperature fluctuations, sand ingress, and accelerated equipment degradation. Practical experience from comparable initiatives, such as the Sahara Solar Plan (Desertec),[49] underscores the criticality of these factors; specifically, historical hurdles regarding water scarcity for cleaning/cooling and the logistical complexity of remote maintenance highlight that theoretical resource abundance does not automatically guarantee implementation success. A highly actionable extension of this work would be the development of an integrated modeling framework that explicitly incorporates the logistics and levelized cost of water procurement (e.g., desalination and long-distance pipeline transport versus local groundwater extraction) directly into the techno-economic optimization of the WSS system. To address these uncertainties within our modeling

Table 1. Comparative analysis of this study versus existing literature on AI energy demand and desert renewable systems

Study/source	Spatial resolution	Temporal resolution	System integration	Key focus
IEA (2025) Reports on AI Energy Demand[5]	regional/national aggregate estimates; no grid-cell specificity	annual demand projections; no hourly modeling	single-sector (AI electricity demand); no renewable generation coupling	macro-scale demand forecasting for policy awareness
Goldman Sachs (2024) AI Era Analysis[46]	market-level aggregation; country/continent scale	annual demand projections; no hourly modeling	quantitative analysis of compute demand and energy efficiency trajectories	power growth drivers and green investment opportunity
de Vries (2023, 2026) on AI sustainability challenges[47,48]	global aggregate; conceptual geographic distribution	not applicable (conceptual framework)	qualitative discussion of energy-water-carbon nexus; no optimization model	identifying sustainability risks and policy imperatives
Previous desert solar studies (Li et al.[42]; Komoto et al.[39])	regional climate models at several 100-km resolution	seasonal/monthly; no sub-hourly dispatch	limited storage consideration; grid export assumption	climate feedback effects and ecological restoration potential
This study	$0.25^\circ \times 0.25^\circ$ grid cells ($\sim$28 km at equator); 40,000 cells assessed globally	hourly optimization over 8,760 h/year with full chronological dispatch modeling	integrated wind-solar-storage (WSS) optimization with data center thermal constraints and latency requirements	end-to-end techno-economic feasibility of off-grid desert AI data centers with 24/7 reliability guarantee

constraints, we conducted a multidimensional sensitivity analysis—incorporating stress tests such as shortening asset lifespans to 15 years, quadrupling maintenance costs, and enforcing substantial storage redundancies (up to 2.0×). Our results indicate that the sheer abundance and high quality of renewable resources provide a sufficient economic buffer to absorb these inefficiencies; however, detailed engineering optimization remains essential for subsequent site-specific implementation.

Second, regarding cooling and water constraints, our economic analysis relies on direct air-cooling systems to minimize water withdrawal. While our modeling based on hourly ERA5 meteorological data confirms that the annual average PUE remains within competitive limits (typically <1.6) across most selected regions, we acknowledge that instantaneous thermal loads during extreme heat events may spike. Furthermore, although operational water consumption is negated, the logistical challenge of sourcing water for periodic maintenance (e.g., cleaning solar panels and heat exchangers) remains a physical constraint that must be managed, potentially through robotic dry-cleaning solutions or distinct supply chains.

Third, our assessment of environmental and climatic suitability adopts conservative but macroscopic criteria. We strictly excluded all protected areas listed in the World Database on Protected Areas (WDPA) to avoid ecological encroachment. However, we recognize that desert ecosystems outside designated reserves can be fragile. Future implementation must be preceded by rigorous, site-specific environmental impact assessments to safeguard local biodiversity. Additionally, while our analysis of climate change scenarios (SSP1-2.6 to SSP5-8.5) suggests that the selected regions will remain climatologically viable through 2100—with potential efficiency losses in photovoltaics potentially offset by reduced aerosol emissions ("brightening")—long-term asset planning must account for these evolving thermal baselines. To make this actionable, future studies could employ advanced Earth system models to generate high-resolution, decadal-scale meteorological projections, enabling dynamic simulations of how climate-driven shifts in wind and solar resources, coupled with rising ambient temperatures, will concurrently impact both the system LCOE and the data center PUE over the asset life cycle.

Fourth, geopolitical and regulatory dimensions are discussed qualitatively. Factors such as land ownership complexities, data sovereignty laws, and cross-border investment risks are decisive for project feasibility but are beyond the scope of this techno-economic model. Future work should prioritize the development of policy frameworks that address these governance barriers. For instance, specific, tractable research could perform comparative institutional analyses to design cross-border energy and data trade agreements that incentivize multinational AIDC investments while protecting national data sovereignty.

Besides, we position the desert AIDC paradigm as one of the sustainable computing futures. Emerging frontiers, such as underwater data centers (UDCs),[50] leveraging natural cooling and offshore energy, or space-based computing infrastructures, offer complementary solutions to terrestrial limitations. The high-resolution spatiotemporal optimization framework established in this work is transferable and can be adapted to evaluate these alternative paradigms, providing a strategic foundation for the global transition toward zero-carbon AI infrastructure. A logical and highly tractable next step would be conducting a comparative spatial analysis of the levelized cost of compute between off-grid desert WSS systems, grid-connected renewables plus storage in specific load-center regions, and emerging UDC models, thereby providing a comprehensive merit-order curve for global AI infrastructure deployment.

MATERIALS AND METHODS

We assess the global potential of desert-based, green computing infrastructure using a spatially resolved modeling framework that couples renewable resource availability with data center operational requirements (Figure 8). This framework synthesizes high-resolution geospatial data processing, power system optimization, and economic sensitivity analysis into a coherent workflow comprising three primary modules. First, the Energy Resource Assessment module integrates multi-source Earth observation data—including land use, topography, and meteorological records from ERA5—to quantify the installed capacity potential and hourly generation profiles of wind and solar resources at a $0.25^\circ \times 0.25^\circ$ spatial resolution. Second, these inputs feed into the Renewable Energy and Storage Combination Optimization (RESCO) model, which determines the cost-optimal configuration of WSS systems required to maintain a target power supply for continuous AIDC operations. Finally, the Environmental and Techno-Economic Analysis module evaluates the deployable capacity and economic viability of these systems by comparing their LCOE against local grid prices, while accounting for critical uncertainties such as PUE and non-IT infrastructure costs.

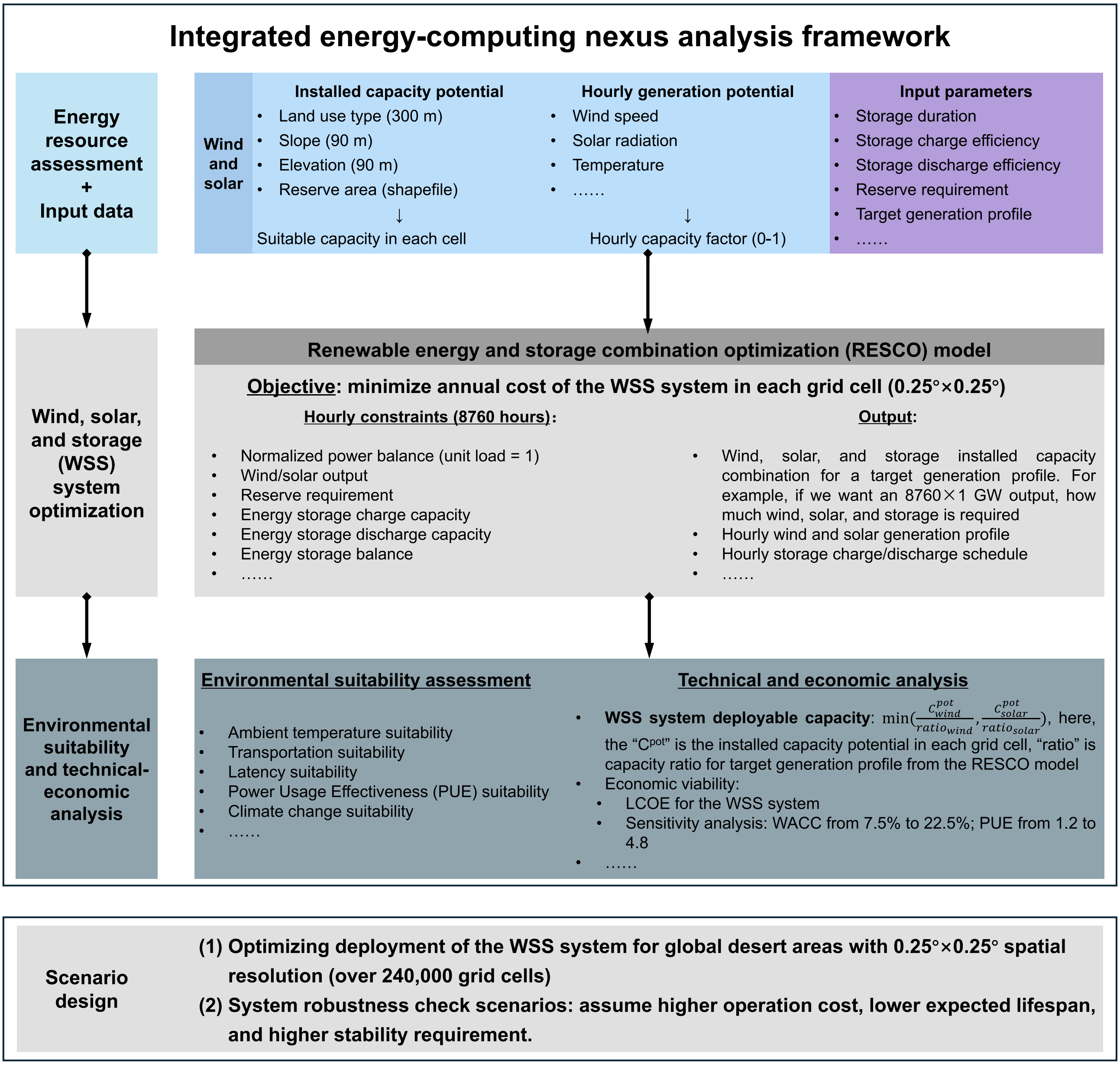

Figure 8. Integrated energy-computing nexus analysis framework developed in this study

Wind and solar resource assessment

This study assesses resources for onshore wind and utility-scale solar PV in the global desert area. We first extract the desert shape files from ESRI[51] with biome label as "Desert & Xeric Shrublands"; see Figure S1. In these desert regions, we estimate the land suitable for the deployment of wind and solar PV according to land use type, slope, altitude, and reserve requirement for biodiversity or natural protection. A global land use dataset from the European Space Agency with 300 × 300 m resolution[52] is used as the base geographic information system layer for further suitability filtering. This dataset divides land area into 22 types according to the criteria released by the Food and Agriculture Organization of the United Nations. Based on this land use layer, we assign the slope and altitude records from refs.[53,54] to each pixel. With the geographical boundary of biodiversity and natural reserve from the UNEP-WCMC and IUCN,[55] we exclude the land use pixels located in these boundaries for the wind and solar deployment (i.e., these pixels are not considered for our WSS systems in this study). We then adopt technical standards to filter out pixels that do not meet engineering requirements (i.e., steep slope or higher altitude). For the remaining pixels, we set a suitability factor (available area to the total pixel area) according to land use and generation type (wind or solar PV) to estimate the suitable area for the WSS systems deployment; see details in Table S2. These areas of each pixel are aggregated to each $0.25^\circ \times 0.25^\circ$ grid cell that is consistent with the ERA5 dataset,[25] which provides comprehensive records of global meteorological data spanning over eight decades. Within the suitable area in each grid cell, we assume a 7 × 7 rotor-diameter spacing for the adjacent wind turbines and fixed optimal tilt for solar PV to estimate the installation density (measured in MW/km^2) to assess the installed capacity potential for each grid cell.[29] See detailed parameters and results for this assessment in sections S1.1–S1.4.

We determine the generation output profile using the hourly capacity factor (0–1, real output to nameplate capacity in a specific time step, e.g., 1 h in this study), which is calculated based on meteorological data related to each renewable type (e.g., wind speed for wind and solar radiation for solar PV). In this study, we adopt the reanalysis dataset from ERA5 in the year 2020 to estimate the hourly generation profile in each 0.25°×0.25° grid cell. For wind power, a curve of power output and wind speed at a 120-m turbine height with model parameters from the National Renewable Energy Laboratory is used to calculate the hourly capacity factor. The ERA5 dataset does not provide 120-m wind speed directly; instead, we use the wind speed data at 10 m and 100 m and the wind profile power law to calculate wind speed at 120-m height. Besides, we also consider a) real air density, using dew-point temperature, pressure, and air humidity; b) wake effects; and c) extreme low temperature and high wind speed at wind turbine height to adjust the capacity factor of our wind resource assessment. For solar PV, we use variables including surface downwelling shortwave radiation (W/m^2), ambient temperature (°C), and 10 m wind speed (m/s) to estimate the capacity factor according to a typical generation model and electric engineering adjustment. See the detailed formula and parameters for wind and solar generation output assessment in the SI section S1.1.

Renewable Energy and Storage Combination Optimization model

To assess the technical feasibility and economic viability of off-grid WSS systems globally, we developed the RESCO model. This spatially resolved linear programming framework determines the optimal capacity configuration of wind, solar, and energy storage to meet a continuous, stable load requirement at the lowest possible cost. The model minimizes the annualized system cost for each grid cell i. The decision variables include the installed capacities of wind (C_{wind}), solar (C_{solar}), and energy storage (C_{st}), as well as the hourly operation variables, including power dispatch ($P_{g,t}$), storage charging/discharging ($P_{\text{ch},t}$,$P_{\text{dis},t}$), and reserve provisions ($R_{g,t}$). The objective function is formulated as follows:

$$\min Z = \sum_{g \in \mathcal{G}} \left(\text{CapEx}_{g,i} \times \text{CRF}_g + \text{FOM}_g\right) \times C_g + \gamma \sum_{t \in \mathcal{T}} \left(P_{\text{ch},t} + P_{\text{dis},t}\right), \qquad \text{(Equation 1)}$$

where Z is the total annualized system cost to be minimized. $\mathcal{G} = \{\text{wind}, \text{solar}, \text{st}\}$ represents the set of technologies. The annualized capital cost for each technology is derived from its specific capital expenditure ($\text{CapEx}_{g,i}$) multiplied by the capital recovery factor (CRF_g), plus the fixed O&M cost (FOM_g). The term γ (set to 10^{-6}) is a sufficiently small penalty factor introduced to prevent simultaneous charging and discharging behaviors of the storage system in the mathematical solver.

In this study, "24/7 stable power supply" is defined as achieving 100% temporal availability, that is, meeting the target load for all 8,760 h of the simulation year ($t \in \{1, \ldots, 8760\}$) without interruption, regardless of meteorological conditions. This is enforced through Equation 2, which mandates exact power balance in every hour t:

$$P_{\text{wind},t} + P_{\text{solar},t} + P_{\text{dis},t} - P_{\text{ch},t} = 1, \forall t, \qquad \text{(Equation 2)}$$

where $P_{\text{wind},t}$ and $P_{\text{solar},t}$ represent the hourly dispatched power from wind and solar PV, respectively. We normalize the target load to unit one to solve for the capacity configuration, which allows for linear scaling based on actual demand. To account for extreme weather events and renewable resource variability, the model further imposes a 5% spinning reserve requirement (r_{req} = 0.05), ensuring the system maintains sufficient dispatchable capacity to respond to unforeseen supply disruptions or load fluctuations (Equation 3).

$$R_{\text{wind},t} + R_{\text{solar},t} + R^{ch}_{\text{st},t} + R^{dis}_{\text{st},t} \geq 1 \times r_{\text{req}}, \forall t. \qquad \text{(Equation 3)}$$

In this formulation, $R_{\text{wind},t}$ and $R_{\text{solar},t}$ denote the spinning reserve provided by the curtailed (spilled) wind and solar power, respectively. The storage system provides reserve through two distinct operational mechanisms, denoted by $R^{ch}_{\text{st},t}$ and $R^{dis}_{\text{st},t}$. Specifically, $R^{ch}_{\text{st},t}$ represents the reserve achieved by reducing the ongoing charging power, while $R^{dis}_{\text{st},t}$ denotes the reserve provided by increasing the discharging power.

Power generation and reserve provision from renewables are physically constrained by the hourly capacity factors ($CF_{g,t}$, ranging from 0 to 1) derived from our resource assessment:

$$P_{g,t} + R_{g,t} \leq C_g \times CF_{g,t}, \forall g \in \{\text{wind}, \text{solar}\}, \forall t. \qquad \text{(Equation 4)}$$

Equation 4 ensures that the sum of the dispatched power and the reserved capacity for any renewable source g does not exceed its available theoretical output at hour t.

Energy storage is modeled using a reservoir logic. The state of charge (E_t), representing the total energy stored in the battery at hour t, evolves based on charging/discharging activities, round-trip efficiencies (η_{ch},η_{dis}), and the time step duration (Δt = 1 h):

$$E_t = E_{t-1} + \left(P_{\text{ch},t} \times \eta_{\text{ch}} - P_{\text{dis},t} / \eta_{\text{dis}}\right) \times \Delta t, \forall t, \qquad \text{(Equation 5)}$$

$$P_{\text{ch},t} \leq C_{\text{st}}; P_{\text{dis},t} \leq C_{\text{st}}; E_t \leq C_{\text{st}} \times \mathcal{D}_{\text{st}}, \forall t. \qquad \text{(Equation 6)}$$

We assume a lithium-ion battery system with a specific storage duration ($\mathcal{D}_{\text{st}}$) of 4 h,[29] meaning the maximum energy capacity is inherently defined as $C_{\text{st}} \times \mathcal{D}_{\text{st}}$. The charging and discharging efficiencies (η_{ch},η_{dis}) are parameterized based on current practice (e.g., 90.25% round-trip efficiency[37,56]). Constraints on storage reserves (Equation 7) ensure that the battery can physically support the committed up/down regulation:

$$R^{ch}_{\text{st},t} \leq P_{\text{ch},t}; R^{dis}_{\text{st},t} \leq C_{\text{st}} - P_{\text{dis},t}, \forall t. \qquad \text{(Equation 7)}$$

As explicitly bounded here, the capability to provide reserve by reducing charging ($R^{ch}_{\text{st},t}$) is limited by the charging power $P_{\text{ch},t}$. Conversely, the reserve provided by increasing discharge ($R^{dis}_{\text{st},t}$) is bounded by the remaining available discharge capacity (i.e., the theoretical maximum discharge capability minus the ongoing discharge power $P_{\text{dis},t}$). Furthermore, to maintain physical feasibility, the storage system's capability to provide reserve is dynamically bounded by its state of charge, ensuring that reserve power is exclusively committed when sufficient stored energy is available.

Economic analysis of the WSS systems for AIDCs

In this study, we assume the investment cost for the IT equipment in a data center within a specific country (e.g., the United States) is the same whether it is deployed in a desert area or not. The investment cost for the non-IT equipment and the PUE are the two primary factors influencing the decision whether it is economical to deploy the data center to the desert area for cheaper electricity. A grid cell has economic potential for building an AIDC if the cost differences from a cheaper electricity supply compared to a non-desert area (e.g., a city) are larger than the cost increase due to operation in a desert area with a required WACC. We adopt a real-world project as the base case (see section S1.2) and design three WACC requirement scenarios (7.5%, 15.0%, and 22.5%) to iterate the rates of non-IT equipment and PUE to the base case, which consist of 1.0× to 4.0× with 0.01× as a step for each type (thus 90,000 combinations) to analyze the uncertainties of cost-effectiveness. This broad WACC range is selected to capture diverse financing environments, where the lower bound aligns with IEA benchmarks for mature utility-scale renewables,[57] while the upper bound reflects the higher risk premiums typical of off-grid, private-sector digital infrastructure. We first estimate the annual cost differences for using electricity in the desert area in each grid cell as follows:

$$\widehat{cost}_{ele} = P_{IT} \times \widehat{PUE} \times lr \times 8760 \times pri_{ele}, \qquad \text{(Equation 8)}$$

$$cost^{i}_{ele} = P_{IT} \times \widehat{PUE} \times pr \times lr \times 8760 \times LCOE^{i}_{WSS}, \qquad \text{(Equation 9)}$$

$$\Delta cost^{i}_{ele} = \widehat{cost}_{ele} - cost^{i}_{ele} \qquad \text{(Equation 10)}$$

$$= P_{IT} \times \widehat{PUE} \times lr \times 8760 \times \left(pri_{ele} - pr \times LCOE^{i}_{WSS}\right),$$

where P_{IT} is the electric capacity (MW) of IT equipment in the data center; $\widehat{PUE}$ is the PUE in the base case (1.2); lr is the average load rate for IT equipment in a year (75% used in our analysis for higher data center utilization in the AI era)[58]; and $P_{IT} \times \widehat{PUE} \times lr \times 8760$ means the annual electricity consumption in the base case; pri_{ele} is the electricity price from local power grid (see Table S8); and $\widehat{cost}_{ele}$ is the electricity cost for an AIDC operating as usually. In Equation 9, pr is the PUE ratio (from 1.0 to 4.0) to the base case (i.e., $pr\times \widehat{PUE}$ is our test PUE in the desert area); $LCOE^{i}_{WSS}$ is the LCOE of the WSS systems in grid cell i; and $cos\ t^{i}_{ele}$ is the electricity cost for an AIDC operating in the desert grid cell i. Therefore, $\widehat{cost}_{ele} - cos\ t^{i}_{ele}$ is the annual cost differences for an AIDC in a desert area for cheaper electricity access. The annual cost increase for non-IT equipment includes two parts: one is the annual capital expenditure for the investment, and the other is the O&M cost, as follows:

$$\widehat{cost}_{fix} = \widehat{CapEx} \times CRF + FOM, \quad \text{(Equation 11)}$$

$$cost_{fix} = \widehat{CapEx} \times cr \times CRF + 1.5 \times FOM, \quad \text{(Equation 12)}$$

$$\Delta cost_{fix} = cost_{fix} - \widehat{cost}_{fix} \quad \text{(Equation 13)}$$

$$= \widehat{CapEx} \times CRF \times (cr - 1) + 0.5 \times FOM,$$

where $\widehat{CapEx}$ is the investment cost for non-IT equipment in the base case; $CRF = \frac{wacc\times(1+wacc)^n}{(1+wacc)^n - 1}$ is the capital recovery factor with a given WACC and operating years (20 years for the non-IT equipment[59]); cr is the CapEx ratio (from 1.0 to 4.0) to the base case (i.e., $cr \times CapEx$ is our test non-IT CapEx in the desert area); FOM is the annual fixed O&M, where we assume this cost in a desert area is 50% larger than that operating as usual. Therefore, $\Delta cost_{fix}$ is our estimated annual cost increase for an AIDC operating in a desert area.
Finally, if the cost differences in a grid cell are larger than the cost increase under a specific $\langle pr,cr \rangle$ scenario, we have the following:

$$\Delta cos\ t^{i}_{ele} - \Delta cos\ t_{fix} = P_{IT} \times \widehat{PUE} \times lr \times 8760 \times \left(pri_{ele} - pr \times LCOE^{i}_{WSS}\right)$$

$$- \widehat{CapEx} \times CRF \times (cr - 1) - 0.5 \times FOM > 0. \quad \text{(Equation 14)}$$

We consider it to have economic potential and add the electricity generation in this grid cell as the economic potential for AIDC operation.

Investment cost assumptions and scenario design

Our economic analysis is based on region-specific investment costs for renewable energy infrastructure in global desert areas. We delineate eight regions for data collection: Africa, Asia (excluding China and India), China, Europe, India, North America, Oceania, and South America. Capital expenditures are estimated for onshore wind, solar PV, and battery storage (see Table S6 for regional values). The respective CapEx ranges are 600–1,200 \$/kW for wind, 486–1,080 \$/kW for solar PV, and 286–1,358 \$/kW for storage.[60] The storage component is assumed to be a lithium-ion battery system with a 4-h duration and 90.25% round-trip efficiency[37,56] (Table S5). To anchor the analysis, we select the 29-MW Guojin Data Center as a real-world base case. To explore future uncertainties, we develop a matrix of scenarios combining three 2030 technology cost projections (CapEx-high, -mid, and -low) with three financial assumptions for the weighted average cost of capital: 7.5%,[57,61] 15.0%, and 22.5% (the higher WACCs represent the higher investment return expectation in high-tech fields).

Reference case: Nanfang Wanguo Guojin Data Center project

Because it is very difficult to get the real financial parameters about data centers for business secrecy, we select a representative project in China with detailed operation data for its prospectus for real estate investment trusts as our reference case.[36] This facility, the Guojin Data Center (GJDC), is operated by Nanfang Wanguo Data Center, a major provider of data center colocation services in China. Located in Kunshan, Jiangsu Province, the GJDC comprises two buildings with four levels above ground and one below, dedicated to housing computational infrastructure. The facility has a gross floor area of 35,376 m^2 and a total designed power capacity of about 30 MW, accommodating 4,192 server racks.

The total capital investment for the GJDC project stands at 823.69 million CNY (approximately 117.67 million USD). The annual FOM costs, exclusive of electricity, represent approximately 0.65% of this initial investment. Using the GJDC as a reference, our model assumes that the fixed O&M costs for a data center operating in a desert environment are 1.5 times this baseline. Furthermore, to comprehensively evaluate the economic landscape, we conduct a sensitivity analysis across a matrix of 90,000 cost combinations for each grid cell. This analysis systematically varies two key parameters—PUE and non-IT capital investment—from 1.0 to 4.0 times their respective baseline values (a PUE of 1.2 and the GJDC investment cost) in increments of 0.01.

While our reference project is situated in China, where non-IT equipment investment costs may be lower than in other regions (e.g., Europe and the United States), our uncertainty analysis rigorously accounts for this variability. Specifically, the multiplier applied to non-IT equipment investment costs could be disaggregated into two distinct components (i.e., $cr = cr_{\alpha}\times cr_{\beta}$ in Equation 12). The first component (cr_{α}) quantifies the cost escalation attributable to a shift from the Chinese baseline to other geographical regions, encompassing factors such as differences in labor rates, material costs, and logistical complexities. The second component (cr_{β}) addresses the incremental cost increase associated with deploying projects in desert environments compared to conventional locations. Industry benchmarks indicate that globally standardized mechanical and electrical equipment accounts for 55%–65% of non-IT capital expenditures, resulting in actual regional cost premiums typically ranging between 1.5× and 2.0× relative to the reference case.[62,63] Consequently, our 4.0× upper bound represents a conservative worst-case scenario that sufficiently accommodates extreme supply chain or labor constraints in regions like the Middle East and Africa.

DATA AND CODE AVAILABILITY

The source code for the optimization model (RESCO) and figure plotting, along with the primary optimization results, are publicly available on GitHub at https://github.com/mrziheng/DesertPowerAIDC.[64] Due to storage size constraints, the raw high-resolution wind and solar capacity factor datasets (exceeding tens of gigabytes) are not hosted in the repository; these specific large-scale datasets are available from the lead contact upon reasonable request.

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AUTHOR CONTRIBUTIONS

Z.Z.: conceptualization, methodology, investigation, software, data collection and calibration, formal analysis, visualization, writing–original draft preparation, and writing-review & editing; R.Y.: conceptualization, data collection and calibration, formal analysis, writing–original draft preparation, and writing-review & editing; W.-Q.C.: writing-review & editing; D.Z.: writing-review & editing, funding acquisition, and supervision.

DECLARATION OF INTERESTS

There are no competing interests to declare.

SUPPLEMENTAL INFORMATION

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